Aimin Li

Postdoctoral Researcher, KTH Royal Institute of Technology
Division of Information Science and Engineering
Stockholm, Sweden | Born: July 1998
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Research Interests

Information Theory, Age of Information, Goal-Oriented and Semantic Communications, Optimal Sampling and Scheduling, Finite-Blocklength Channel Coding, and Non-Terrestrial and Space Networks

Education

Harbin Institute of Technology, Shenzhen *Sep 2020 – 2025*
Project 985, China
Ph.D., *Information and Communication Engineering* **GPA rank: 1st**
Thesis Title: "**Key Technologies for Timeliness-Enhanced Space-Air Communications in Perception-Decision Closed-Loop Systems**"
Advisor: **Prof. Shaohua Wu**

Harbin Institute of Technology, Shenzhen *Aug 2016 – Jun 2020*
Project 985, China
B.S., *Electrical Engineering* **GPA rank: 2nd**
Thesis Title: "**Performance Analysis and Encoding Algorithm Design of Spinal Codes in the Finite Blocklength Regime**"
Advisor: **Prof. Shaohua Wu**

Research Experience

Postdoctoral Researcher, KTH Royal Institute of Technology, Stockholm *Oct 2026 – Present*
Division of Information Science and Engineering. Advisor: **Prof. Mikael Skoglund** (IEEE Fellow).

- Swedish Research Council (VR) project on joint communication and decision-making for distributed systems.

Postdoctoral Researcher, Middle East Technical University, Ankara *Oct 2025 – Oct 2026*
Communication Networks Research Group (CNG). Advisor: **Prof. Elif Uysal** (IEEE Fellow).

- Researcher on the **ERC Advanced Grant GO-SPACE**, jointly with the German Aerospace Center (DLR).
- Developed age-optimal preemption theory under heavy-tailed delays [C18] and goal-oriented designs for interplanetary and non-terrestrial networks [J24].
- Co-supervised 11 B.S., M.S., and Ph.D. students, leading to papers at ISIT, ITW, MobiHoc, Asilomar, WiSEE, and IAC.

Visiting Researcher, A*STAR, Singapore *Oct 2023 – Oct 2024*
Institute for Infocomm Research (I^2R). Advisor: **Prof. Sumei Sun** (IEEE Fellow).

- Built a goal-oriented sampling framework for remote decision making [J16], [C20].
- Derived tight finite-blocklength error bounds for Spinal codes [J17], [J18], [J19].

Awards and Honors

Best Dissertation Award Nomination *2025*
Harbin Institute of Technology (HIT), China. **Sole nominee** of the EE Department and one of only two nominees in the School.

Provincial Outstanding Graduate Award *2025*
Heilongjiang Province, China. Provincial-level honor for graduates, awarded to the **top 0.4%** in the province.

Outstanding Graduate Student Award *2025*
Harbin Institute of Technology (HIT), China. Awarded to the top 10% of graduates.

3rd Prize, IEEE GLOBECOM 4MT Thesis Competition *Dec 2024*
IEEE Communications Society's worldwide doctoral thesis competition. Ranked **5th among 15 finalists worldwide**; presented by the IEEE ComSoc President.

IEEE GLOBECOM Student Travel Grant *Dec 2024*
Granted to the **top 15 Ph.D. students worldwide** selected for the 4MT finals.

National Scholarship of China (twice) *2020, 2024*
The highest-level government scholarship for graduate students in China, awarded to the **top 0.2%** nationwide.

NSFC Young Students Basic Research Program (Ph.D. Students) *2024*
National Natural Science Foundation of China (NSFC). Selected candidate in the **first national cohort**; fewer than 600 Ph.D. students across all disciplines in China, and the sole Ph.D. student of the EE Department.

HIT "Dianzi" Research Student Program *2023*
Harbin Institute of Technology, China. University-level independent PI research project; **sole Ph.D. recipient** of the EE Department.

China Scholarship Council (CSC) Scholarship *2023*
Chinese government scholarship supporting visiting research at A*STAR, Singapore.

Special Merit Doctoral Scholarship *2020*
Harbin Institute of Technology, China. The highest university scholarship for incoming Ph.D. students, awarded by the University President to **only 8 Ph.D. students university-wide** (sole recipient across 13 departments at HIT Shenzhen).

Best Undergraduate Thesis Award *2020*
Harbin Institute of Technology (HIT), China. **One of two recipients** in the EE Department.

Outstanding Participant, Tsinghua University Graduate Summer Camp *2019*
Tsinghua University, China. Qualified for postgraduate admission.

Gongjin Scholarship for Overall Excellence *2018*
Harbin Institute of Technology, China. Awarded by the University Vice President to **only 3 students** of the EE Department.

Provincial Merit Student *2018*
Heilongjiang Province, China. Awarded to the **top 1%** of students in the province.

First Prize, Chinese Mathematics Competitions (twice) *2016, 2017*
China-wide undergraduate mathematics competition organized by the Chinese Mathematical Society.

Outstanding Student Award (twice) and First-Class Scholarship *2017, 2019*
Harbin Institute of Technology (HIT), China. Top 10% (Outstanding Student) and top 5% (First-Class Scholarship).

PUBLICATIONS

Aimin Li in bold; underlined authors are students I mentored; # co-first authors; † corresponding author.

Journals

[J1]: *ASTRA: Asynchronous Age-Aware Satellite Random Access with Correlated Multi-Satellite Fusion.*
S. Chakraborty, **Aimin Li**†, Y. İnce, S. Baghaee, and E. Uysal; **under review at** IEEE Transactions on Wireless Communications, 2026.

[J2]: *Age-Optimal Sampling and Routing under Intermittent Links and Energy Constraints.*
A. U. Atasayar, **Aimin Li**†, Ç. Arı, and E. Uysal; **under review at** IEEE Transactions on Mobile Computing, 2026.

[J3]: *Source PAC Coding for Low-Latency Secret Key Generation in the Short-Blocklength Regime.*
L. Song, D. Zhang, **Aimin Li**, and T. Zhang; **under review at** IEEE Transactions on Information Theory, 2025.

[J4]: *Multi-Agent Diffusion Planning with Stable Online Reinforcement Fine-Tuning for Cooperative Driving.*
H. Bai#, **Aimin Li**#, R. Yao, X. Zhao, T. Zhang, X. Zhang, C. Li, L. Gao, and J. Ma; **under review at** IEEE Transactions on Mobile Computing, 2026.

[J5]: *Permanent Connection Guaranteed Resilient LEO Constellation for Remote Area Communications.*
G. Chen, Y. Deng, **Aimin Li**, and S. Wu; IEEE Transactions on Wireless Communications, 2026.

[J6]: *AoI-Aware Joint Sampling-Buffering-Routing Optimization for Autonomous UAV Swarms via a MARL Approach.*
H. Wu, S. Wu, **Aimin Li**, S. Meng, and Q. Zhang; IEEE Transactions on Mobile Computing, 2026.

[J7]: *Entropy-Driven Sampling for Remote Estimation in Internet of Things Systems.*
X. Chen, **Aimin Li**, Y. Deng, and S. Wu; IEEE Internet of Things Journal, 2026.

[J8]: *Information Freshness and Timeliness Analysis in the Finite Blocklength Regime for Mission-Critical Applications.*
D. Zhang, M. Sun, L. Song, S. Jia, **Aimin Li**, T. Zhang, A. Al-Dulaimi, and S. Mumtaz; IEEE Transactions on Communications, 2025.

[J9]: *MARHO: Hybrid Task Offloading in Maritime MEC via Multi-Agent Reinforcement Learning.*
J. Ning, **Aimin Li**, G. C. F. Lee, S. Sun, and T. Yang; IEEE Open Journal of the Communications Society, 2025.

[J10]: *Effectiveness-Oriented SAGSIN: Unveiling a Unified Metric and a Comprehensive Framework.*
S. Meng, S. Wu, H. Wu, **Aimin Li**, and Q. Zhang; IEEE Network, 2024.

[J11]: *Toward Goal-Oriented Semantic Communications: AoII Analysis of Coded Status Update System under FBL Regime.*
S. Meng, S. Wu, **Aimin Li**, and Q. Zhang; IEEE Journal on Selected Areas in Information Theory, 2023.

[J12]: *Analysis and Optimization of the HARQ-Based Spinal Coded Timely Status Update System.*
S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang; IEEE Transactions on Communications, 2022.

[J13]: *Partial Self-Concatenation Structure and Performance Analysis of Spinal Codes over Rayleigh Fading Channel.*
S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang; IEEE Transactions on Vehicular Technology, 2022.

[J14]: *Minimizing Age-of-Information in HARQ-CC Aided NOMA Systems.*
S. Wu, Z. Deng, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang; IEEE Transactions on Wireless Communications, 2022.

[J15]: *A Weighted Graph-Based Handover Strategy for Aeronautical Traffic in LEO SatCom Networks.*
X. Lv, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang; IEEE Networking Letters, 2022.

[J16]: *From Freshness to Effectiveness: Goal-Oriented Sampling for Remote Decision Making.*
Aimin Li, S. Wu, G. C. F. Lee, and S. Sun; IEEE Transactions on Information Theory, 2026.

[J17]: *Unified Upper Bounds on the ML Decoding Error Probability of Spinal Codes over Fading Channels.*
Aimin Li, X. Chen, S. Wu, G. C. F. Lee, and S. Sun; IEEE Transactions on Wireless Communications, 2026.

[J18]: *Error Floor of ML-Decoded Spinal Codes in the Finite Blocklength Regime.*
Aimin Li, S. Wu, X. Chen, and S. Sun; IEEE Transactions on Vehicular Technology, 2025.

[J19]: *Tight Upper Bounds on the BLER of Spinal Codes over the AWGN Channel.*
Aimin Li, S. Wu, X. Chen, and S. Sun; IEEE Transactions on Communications, 2025.

[J20]: *Toward Goal-Oriented Semantic Communications: New Metrics, Framework, and Open Challenges.*
Aimin Li, S. Wu, S. Meng, R. Lu, S. Sun, and Q. Zhang; IEEE Wireless Communications, 2024.

[J21]: *Goal-Oriented Tensor: Beyond Age of Information Toward Semantics-Empowered Goal-Oriented Communications.*
Aimin Li, S. Wu, S. Sun, and J. Cao; IEEE Transactions on Communications, 2024.

[J22]: *Age of Information with Hybrid-ARQ: A Unified Explicit Result.*
Aimin Li, S. Wu, J. Jiao, N. Zhang, and Q. Zhang; IEEE Transactions on Communications, 2022.

[J23]: *Spinal Codes over Fading Channel: Error Probability Analysis and Encoding Structure Improvement.*
Aimin Li, S. Wu, J. Jiao, N. Zhang, and Q. Zhang; IEEE Transactions on Wireless Communications, 2021.

[J24]: *Goal-Oriented Communications for Interplanetary and Non-Terrestrial Networks.*
E. Uysal and **Aimin Li**; IEEE Communications Magazine, 2026.

Conference Proceedings

[C1]: *Worse Routes, Fresher Information: Preemption Redefines Which Route is Best.*
A. U. Atasayar, **Aimin Li**, and E. Uysal; ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc) Workshops, 2026.

[C2]: *On the Multi-Weight Shift-Sum Iterative Decoding of Short BCH Codes.*
Y. Yang, D. Zhang, **Aimin Li**, Y. Shen, A. A. Frolov, and B. Gu; IEEE Global Communications Conference (GLOBECOM), 2026.

[C3]: *Capacity Analysis of Weibull Fading Channels for Satellite-Ground Integrated Communications.*
A. Lei, D. Zhang, **Aimin Li**, and E. Khorov; IEEE Vehicular Technology Conference (VTC2026-Spring), 2026.

[C4]: *Goal-Oriented Bundle Management and Flow Control for Deep-Space Communications via Reinforcement Learning.*
S. Baghaee, **Aimin Li**, S. Balasubramanian, and E. Uysal; International Astronautical Congress (IAC), 2026.

[C5]: *Fresh Data Delivery: Joint Sampling and Routing for Minimizing the Age of Information.*
A. U. Atasayar, **Aimin Li**, Ç. Arı, and E. Uysal; ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing

(MobiHoc), Poster, 2025. **Acceptance rate: 23%.**

[C6]: *Joint Transmission and Control in a Goal-Oriented NOMA Network.*
K. Liu, S. Wu, **Aimin Li**, and Q. Zhang; IEEE International Conference on Communications (ICC), 2025.

[C7]: *Optimal Sampling for Uncertainty-of-Information Minimization in a Remote Monitoring System.*
X. Chen, **Aimin Li**, and S. Wu; IEEE Information Theory Workshop (ITW), 2024.

[C8]: *Packet Management of AoI in the Finite Block-Length Regime.*
M. Sun, D. Zhang, S. Jia, and **Aimin Li**; IEEE International Conference on Communication Technology (ICCT), 2024.

[C9]: *Deep Reinforcement Learning-Assisted Age-Optimal Transmission Policy for HARQ-Aided NOMA Networks.*
K. Liu, **Aimin Li**, and S. Wu; IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), 2023.

[C10]: *Analysis of GEO Satellite Relay Coded Systems.*
J. Zhang, S. Wu, **Aimin Li**, J. Jiao, and Q. Zhang; IEEE Vehicular Technology Conference (VTC2022-Fall), 2022.

[C11]: *Analysis and Optimization of Spinal Codes over the BSC: From the AoI Perspective.*
S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang; IEEE International Conference on Communications Workshops (ICC Workshops), 2021.

[C12]: *ASTRA: Asynchronous Age-Aware Satellite Random Access via Mean-Field Control.*
S. Chakraborty[#], **Aimin Li**[#], Y. İnce, S. Baghaee, and E. Uysal; Asilomar Conference on Signals, Systems, and Computers, 2026.

[C13]: *In-Orbit Intelligence or Ground Offloading? Inference Freshness under Intermittent Satellite Connectivity.*
A. N. Pehlivanoğlu[#], **Aimin Li**[#], E. Uysal, S. Ciftci, and M. Ozdem; IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), 2026.

[C14]: *SCORP: Scene-Consistent Multi-Agent Diffusion Planning with Stable Online Reinforcement Post-Training for Cooperative Driving.*
H. Bai[#], **Aimin Li**[#], R. Yao, X. Zhao, T. Zhang, X. Zhang, L. Gao, and J. Ma; IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

[C15]: *Age of Information in Time-Varying Multi-Priority Queues.*
B. Karasakal[#], **Aimin Li**[#], and E. Uysal; IEEE Information Theory Workshop (ITW), 2026.

[C16]: *Tight Upper Bounds on the Error Probability of Spinal Codes over Fading Channels.*
X. Chen[#], **Aimin Li**[#], and S. Wu; IEEE International Symposium on Information Theory (ISIT), 2023.

[C17]: *Idle Space: Symbiotic Throughput in AoI-Optimal Status Update Systems.*
Aimin Li and E. Uysal; **under review at** IEEE Conference on Computer Communications (INFOCOM), 2027.

[C18]: *Taming the Heavy Tail: Age-Optimal Preemption.*
Aimin Li, Y. İnce, and E. Uysal; IEEE International Symposium on Information Theory (ISIT), 2026.

[C19]: *Optimal Sampling and Scheduling for Remote Fusion Estimation of Correlated Wiener Processes.*
Aimin Li and E. Uysal; Allerton Conference on Communication, Control, and Computing, 2025.

[C20]: *Sampling to Achieve the Goal: An Age-Aware Remote Markov Decision Process.*
Aimin Li, S. Wu, G. C. F. Lee, X. Chen, and S. Sun; IEEE Information Theory Workshop (ITW),

2024.

[C21]: *Goal-Oriented Tensor: Beyond AoI Towards Semantics-Empowered Goal-Oriented Communications.* **Aimin Li**, S. Wu, and S. Sun; IEEE Wireless Communications and Networking Conference (WCNC), 2024.

[C22]: *Analyzing Age Performance of Hybrid-ARQ: A Unified Explicit Result.* **Aimin Li**, S. Wu, Y. Deng, J. Jiao, N. Zhang, and Q. Zhang; IEEE Global Communications Conference (GLOBECOM), 2022.

[C23]: *Spinal Codes over BSC: Error Probability Analysis and the Puncturing Design.* **Aimin Li**, S. Wu, Y. Wang, J. Jiao, and Q. Zhang; IEEE Vehicular Technology Conference (VTC2020-Spring), 2020.

Theses

[T1]: *Key Technologies for Timeliness-Enhanced Space-Air Communications in Perception-Decision Closed-Loop Systems.* **Aimin Li**; Ph.D. Thesis, Harbin Institute of Technology, 2025. **Best Dissertation Award Nomination.**

[T2]: *Performance Analysis and Encoding Algorithm Design of Spinal Codes in the Finite Blocklength Regime.* **Aimin Li**; B.S. Thesis, Harbin Institute of Technology, 2020. **Best Undergraduate Thesis Award.**

Research Projects and Funding

As Principal Investigator

- *HIT "Dianzi" Research Student Program* (independent PI project), Harbin Institute of Technology, 2023.
- *NSFC Young Students Basic Research Program (Ph.D. Students)*, selected candidate, 2024.

As Researcher

- *Swedish Research Council (VR) project on communication and decision-making for distributed systems*, PI: Prof. Mikael Skoglund, KTH, 2026 – present.
- *ERC Advanced Grant GO-SPACE*, PI: Prof. Elif Uysal, METU, jointly with DLR, 2025 – 2026.

Research Leadership and Mentoring

I have initiated and led research collaborations with 22 junior students across three institutions. I shaped the research questions, mathematical models, and solution frameworks, and mentored analysis, simulation, and implementation toward publications and graduate or career placements, including Ph.D. admissions at **KTH**, **Caltech**, and **HKU**. Authorship roles below refer to me.

Middle East Technical University (METU), 2025 – 2026

- **Adem Utku Atasayar** (M.S., 2025–26): co-supervised thesis research on sampling and routing, leading to MobiHoc [C5, C1] (**co-author**) and a TMC submission [J2] (**co-author**); now a Ph.D. student at METU.
- **Burak Karasakal** (B.S., 2025–26): mentored research on multi-priority queues, leading to ITW [C15] (**co-first author**); recipient of the **IEEE ITW Student Travel Grant (2026)** and admitted to the Ph.D. program at **Caltech**.
- **Yiğit İnce** (M.S., 2025–26): mentored research on preemption and satellite random access, leading to ISIT [C18] (**first author**), Asilomar [C12] (**co-first author**), and a TWC submission [J1] (**co-**

author).

- **Ayşe Nur Pehlivanoğlu** (B.S./M.S., 2025–26): mentored research on satellite inference freshness, leading to WiSEE [C13] (**co-first author**).
- **Sayam Chakraborty** (B.S., 2026): mentored thesis research on satellite random access, leading to Asilomar [C12] (**co-first author**) and a TWC submission [J1] (**co-author**); now at ISRO.
- **S. Balasubramanian** (B.S., 2026): mentored thesis research on deep-space communications, leading to IAC [C4] (**co-author**); admitted to the M.S. program at **NUS**.
- **Çağrı Arı** (Ph.D.): collaborated on sampling and routing research, including MobiHoc [C5] (**co-author**); now at Aselsan.
- **Süleyman Çıtır** (M.S.): mentored through research collaboration.
- **Selay Nisa Şahin** (B.S. intern, 2026): supervised a research internship.
- **Yusuf** (B.S. intern, 2026): supervised a research internship.
- **Aral** (B.S. intern, 2026): supervised a research internship.

Harbin Institute of Technology, Shenzhen (HIT), 2020 – 2026

- **Siqi Meng** (B.S./Ph.D., 2020–24): co-supervised undergraduate thesis and doctoral research on timely and goal-oriented communications, leading to WCM, TCOM, TVT, JSAIT, and IEEE Network (**first author**: [J20]; **co-author**: [J12, J13, J11, J10]).
- **Xiaomeng Chen** (B.S./M.S., 2023–26): co-supervised undergraduate and master's theses on coding and remote estimation, leading to TWC, IoTJ, ISIT, and ITW (**first author**: [J17]; **co-author**: [J7, C7]; **co-first author**: [C16]); admitted to the Ph.D. program at **KTH**.
- **Haoxu Wu** (Ph.D., 2025–26): co-supervised research on AoI-aware UAV swarm scheduling, leading to TMC [J6] (**co-author**).
- **Haojie Bai** (Ph.D.): collaborated on research on cooperative driving, leading to IROS [C14] (**co-first author**) and a TMC submission [J4] (**co-first author**); now at Huawei.
- **Kunpeng Liu** (B.S./M.S., 2024–26): co-supervised undergraduate and master's theses on NOMA networks, leading to ICC and INFOCOM Workshops [C6, C9] (**co-author**); now at Huawei.
- **Jiaming Zhang** (B.S./M.S., 2022–25): mentored undergraduate thesis and master's research on satellite relay coded systems, leading to VTC [C10] (**co-author**).
- **Xin Jin** (B.S., 2022): co-supervised the undergraduate thesis.

Zhengzhou University (ZZU), 2025 – 2026

- **Aiwei Lei** (B.S., 2026): mentored research on satellite-ground fading channels, leading to VTC [C3] (**co-author**); admitted to the Ph.D. program at **HKU**.
- **Yinglei Yang** (M.S., 2026): mentored research on short BCH code decoding, leading to GLOBECOM [C2] (**co-author**).
- **Mingxiao Sun** (M.S., 2025): mentored research on finite-blocklength information freshness, leading to TCOM and ICCT [J8, C8] (**co-author**).
- **Lulu Song** (Ph.D., 2026): mentored research on short-blocklength secret key generation, leading to a TIT submission [J3] (**co-author**).

Teaching Experience

Teaching Assistant, Foundations of Information Theory (graduate), HIT Shenzhen *Fall 2022*

- Supported the graduate information theory course taught by Prof. Shaohua Wu.

Teaching Assistant, Communication Theory (undergraduate), HIT Shenzhen *Fall 2020*

- Supported the undergraduate communication theory course taught by Prof. Shaohua Wu.

SERVICE

- Workshop Co-Chair, IEEE ICC 2027 Workshop on *World Models for 6G-Native Networked Robotics*
- Guest Editor, *Entropy* (MDPI) Special Issue "Information Freshness and Semantic Communications", 2026 – 2027
- Guest Editor, *Electronics* (MDPI), 2026
- Session Chair for the Age of Information session at IEEE ITW 2024
- Session Chair for the Distributed Learning session at IEEE GLOBECOM 2024
- Member of the TPC at ICFEEIE 2024
- Volunteer, IEEE 6G Summit 2023, Singapore
- Volunteer, IEEE 6G Summit 2024, Singapore
- Volunteer, IEEE VTC2024-Spring, Singapore
- Reviewer for the following journals
  - IEEE Transactions on Information Theory
  - IEEE Journal on Selected Areas in Communications
  - IEEE/ACM Transactions on Networking
  - IEEE Transactions on Mobile Computing
  - IEEE Transactions on Communications
  - IEEE Transactions on Wireless Communications
  - IEEE Transactions on Neural Networks and Learning Systems
  - IEEE Transactions on Vehicular Technology
  - IEEE Transactions on Cognitive Communications and Networking
  - IEEE Internet of Things Journal
  - IEEE Communications Letters
  - IEEE Wireless Communications Letters
  - *Engineering*
  - *Scientific Reports*
  - Elsevier Knowledge-Based Systems
  - Elsevier Computer Networks
  - Elsevier Computer Communications
  - Elsevier Ad Hoc Networks
  - AEÜ International Journal of Electronics and Communications
  - MDPI *Electronics*
- Reviewer for the following conferences
  - IEEE ISIT 2023, 2026
  - IEEE ITW 2026
  - IEEE INFOCOM 2024
  - IEEE GLOBECOM 2022, 2023, 2024, 2025, 2026
  - IEEE WCNC 2022, 2024, 2026
  - IEEE VTC 2022
  - IEEE ICCC 2023
  - IEEE ICMLCN 2024
  - IEEE APWCS 2026
  - CSAE 2023
  - ICFEEIE 2024

Talks

Taming the Heavy Tail: Age-Optimal Preemption

- IEEE ISIT 2026, Guangzhou, China *Jul 2026*
- Research Seminar, Sun Yat-sen University, Shenzhen, China *Jul 2026*

4MT Thesis Presentation, IEEE GLOBECOM 2024, Cape Town, South Africa
3rd Prize, IEEE GLOBECOM 4MT Thesis Competition *Dec 2024*

Sampling to Achieve the Goal: An Age-Aware Remote Markov Decision Process
IEEE ITW 2024, Shenzhen, China *Nov 2024*

Goal-Oriented Tensor: Beyond AoI Towards Semantics-Empowered Goal-Oriented Communications
IEEE WCNC 2024, Dubai, UAE *Apr 2024*

Sampling and Scheduling for Goal-Oriented Communications
Research seminar jointly organized by Harbin Institute of Technology and Peng Cheng Laboratory (PCL), Shenzhen, China *May 2023*

Analyzing Age Performance of Hybrid-ARQ: A Unified Explicit Result
IEEE GLOBECOM 2022, Rio de Janeiro, Brazil, *Online* *Dec 2022*

Spinal Codes over BSC: Error Probability Analysis and the Puncturing Design
IEEE VTC2020-Spring, Antwerp, Belgium, *Online* *May 2020*